

Pendubot Control Scheme Based on Nonlinear MPC and MHE Exploiting Parallelization

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Abstract—In this paper we present a real-time optimal control scheme of a Pendubot based on nonlinear model predictive control (NMPC) combined with nonlinear moving horizon estimation (NMHE). For the control of this fast, under-actuated nonlinear mechatronic system we utilize the ACADO Code Generation tool to obtain a highly efficient Gauss-Newton real-time iteration algorithm tailored for solving the underlying nonlinear optimization problems. To further improve the solvers' performance, we aim to parallelize particular algorithmic tasks within the estimation-control scheme. The overall control performance is experimentally verified by steering the Pendubot into its top unstable equilibrium. We also provide a computational efficiency analysis addressing different hardware/software configurations.

I. INTRODUCTION

The double inverted Pendubot, introduced in [1], is a well-known academic benchmark from the class of under-actuated mechatronic systems. During the past decades, these systems have been studied in many application areas, and tend to be used to demonstrate various concepts in linear and nonlinear control such as swing-up, stabilization or trajectory tracking of unstable systems; see e.g. [2], [3]. With regards to process nonlinearities, constraints and stability guarantees, and the dynamic coupling between the unactuated state variables and the actuated ones, optimization-based techniques such as model predictive control can be thus sought as a systematic methodology to tackle these challenging problems. The Pendubot, as a typical nonlinear under-actuated mechatronic system with fast dynamics, may clearly benefit from a complex control strategy capable of addressing the aforementioned control problems and phenomena.

When targeting fast systems and processes, linear MPC has become widely and well-established in both academia and industry due to the recent development of efficient convex programming solvers. Similarly, in case of fast nonlinear mechatronic systems, as is the Pendubot, an increasing interest is in achieving reliable and accurate performance over the widest operating range possible while respecting the imposed constraints. In such control scenarios, instead of re-linearizing the system dynamics, the related optimal control problem (OCP) can be instead cast as a nonlinear MPC problem. Its non-convex nature and inherent convergence issues, however, typically come at price of fairly higher computational requirements for effective solution, as compared to linear formulations. The applications of NMPC were therefore primarily with systems and processes exhibiting rather slower dynamics. This was a motivation for several approaches to tackle the control

of fast, nonlinear and constrained systems, which have been proposed over the last decade, among them e.g. the advanced step controller, the continuation/GMRES method, and the real-time iteration scheme. A comprehensive overview of nonlinear optimization methods can be found in [4], [5].

Within this study we propose an NMHE-based nonlinear MPC control scheme to achieve the Pendubot's top unstable equilibrium. To achieve this challenging objective under real-time conditions we make use of the real-time iteration (RTI) scheme of [6] exploiting the method of direct multiple shooting ([7]). In particular, to avoid the often restrictive computational times typically arising in conventional NMPC/NMHE utilize an efficient customized source code generated by means of the ACADO toolkit of [8], namely its Code Generation tool ([9]). The efficacy of its backbone RTI scheme stems primarily from performing only one Newton-type sequential quadratic programming (SQP) iteration per sampling instant ([10]). In contrast to the plethora of studies, wherein the control of pendulum-like systems is usually addressed by a switching swing-up/balancing strategy, the investigated NMPC scheme exploits the full nonlinear dynamics and thus allows for more complex control tasks within one unified and continuous (no controller switching) control strategy. As outlined above, the present control framework is reasonably augmented by the complementary nonlinear MHE scheme to estimate the unmeasured states of the system. To access the full potential of the RTI scheme, we propose to parallelize its certain algorithmic components. This also allows to exploit the computational possibilities of multicore embedded control platforms. We illustrate this idea within a detailed timing analysis for the investigated control objective.

Furthermore, unlike in majority of similar works, we consider a nonlinear friction model of [11] including both static and dynamic friction. In this way, the modeling error can be significantly mitigated, which directly reflects in the improved estimation-control performance without any commonly arising unexpected behaviour. This phenomenon as well as the overall performance of the proposed NMHE-NMPC scheme is verified experimentally on a real laboratory device. This work has been motivated by the preliminary simulation results obtained in [12]. The indicated practical applicability of the scheme was experimentally verified in [13], however, lacking a more accurate friction model, and further pointing to the possibility of exploiting parallelization. Both of these aforementioned aspects are addressed in this study.

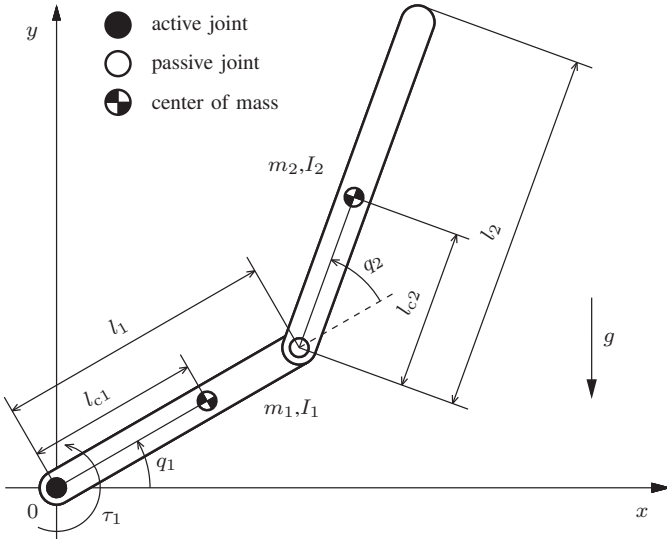


Fig. 1. Schematic of the Pendubot in a relative coordinate system.

II. SYSTEM DYNAMICS

The Pendubot is a two-link planar robot belonging to the class of underactuated mechanical systems since the number of its degrees of freedom is greater than the number of control inputs. The actuator is placed at the base of the first link while the unactuated joint between the two links allows the second link to move freely. Its mathematical model can be derived by means of the Lagrange formalism

$$\frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} \mathcal{L}(q, \dot{q}) - \frac{\partial}{\partial q_k} \mathcal{L}(q, \dot{q}) = \tau \quad k = 1, 2 \quad (1)$$

with the Lagrange function $\mathcal{L}(q, \dot{q}) = \mathcal{T}(q, \dot{q}) - \mathcal{V}(q)$ defined as the difference between the kinetic energy and the potential energy, respectively. The generalized coordinates summarized in the vector $q = [q_1, q_2]^T$ here stand for the angular positions of the two links, and $\tau = [\tau_1, 0]^T$ denotes the external control force vector.

By applying (1), the resulting equation of motion can be cast in the standard vector/matrix form:

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau \quad (2)$$

$D(q)$ denotes the symmetric positive-definite inertia matrix, the Coriolis and centrifugal terms are summarized in $C(q, \dot{q})$ and $g(q)$ is the vector of gravitational terms. For the Pendubot system, schematically illustrated in Fig. 1, the following quantities can be obtained:

$$D(q) = \begin{bmatrix} \theta_1 + \theta_2 + 2\theta_3 \cos q_2 & \theta_2 + \theta_3 \cos q_2 \\ \theta_2 + \theta_3 \cos q_2 & \theta_2 \end{bmatrix},$$

$$C(q, \dot{q}) = \begin{bmatrix} -\theta_3 \sin(q_2) \dot{q}_2 & -\theta_3 \sin(q_2) \dot{q}_2 - \theta_3 \sin(q_2) \dot{q}_1 \\ \theta_3 \sin(q_2) \dot{q}_1 & 0 \end{bmatrix},$$

$$g(q) = \begin{bmatrix} \theta_4 g \cos q_1 + \theta_5 g \cos(q_1 + q_2) \\ \theta_5 g \cos(q_1 + q_2) \end{bmatrix},$$

with physical parameters $\theta_1 = m_1 l_{c1}^2 + m_2 l_1^2 + I_1$, $\theta_2 = m_2 l_{c2}^2 + I_2$, $\theta_3 = m_2 l_1 l_{c2}$, $\theta_4 = m_1 l_{c1} + m_2 l_1$, and $\theta_5 = m_2 l_{c2}$.

Based on the information from the sensor manufacturer, the phenomenon of friction in the passive joint of the laboratory

Pendubot model can be assumed negligible. However, from visual observation, significant friction is present in the actuator. Thus considering friction only in the active joint, the friction vector takes the form of $f_r(\dot{q}) = [f_r(\dot{q}_1) 0]^T$. Here, $f_r(\dot{q}_1)$ represents a nonlinear dynamic friction model, proposed in [11], which is given by:

$$f_r(\dot{q}_1) = c_1 (\tanh(c_2 \dot{q}_1) - \tanh(c_3 \dot{q}_1)) + c_4 \tanh(c_5 \dot{q}_1) + c_6 \dot{q}_1 \quad (3)$$

where $c_i, i = 1, \dots, 6$, are unknown positive constant coefficients. This friction model is continuously differentiable and symmetric at the origin. The static friction coefficient is approximated by $c_1 + c_2$, where as the viscous friction is modelled by $c_6 \dot{q}_1$. The friction model also includes the Coulomb friction via the term $c_4 \tanh(c_5 \dot{q}_1)$, while the Stribeck effect is captured by $\tanh(c_2 \dot{q}_1) - \tanh(c_3 \dot{q}_1)$.

Considering the friction model (3), the dynamic system (2) can be rewritten as follows:

$$\ddot{q}_1 = \frac{1}{\theta_1 \theta_2 - \theta_3^2 \cos^2 q_2} \left[\theta_2 \theta_3 \sin q_2 (\dot{q}_1 + \dot{q}_2)^2 + \theta_3^2 \cos q_2 \sin(q_2) \dot{q}_1^2 - \theta_2 \theta_4 g \cos q_1 + \theta_3 \theta_5 g \cos q_2 \cos(q_1 + q_2) + \theta_2 \tau_1 - \theta_2 f_r(\dot{q}_1) \right] \quad (4)$$

$$\ddot{q}_2 = \frac{1}{\theta_1 \theta_2 - \theta_3^2 \cos^2 q_2} \left[-\theta_3 (\theta_2 + \theta_3 \cos q_2) \sin q_2 (\dot{q}_1 + \dot{q}_2)^2 - (\theta_1 + \theta_3 \cos q_2) \theta_3 \sin(q_2) \dot{q}_1^2 + (\theta_2 + \theta_3 \cos q_2) (\theta_4 g \cos q_1 - \tau_1) - (\theta_1 + \theta_3 \cos q_2) \theta_5 g \cos(q_1 + q_2) + (\theta_2 + \theta_3 \cos q_2) f_r(\dot{q}_1) \right] \quad (5)$$

yielding Pendubot's non-linear equations of motion, which are used for design of a nonlinear predictive controller.

We introduce the state vector $x(t) = [x_1, x_2, x_3, x_4, x_5]^T = [q_1, \dot{q}_1, q_2, \dot{q}_2, u]^T$, where, apart from the naturally considered positions and velocities of both links, the system input denoted by $u = \tau_1$ is augmented. The aim here is to explicitly penalize the input rate $u_R = \dot{u}$ within the NMPC formulation, in order to allow for tuning of the aggressiveness of the control action during the transitions between equilibrium positions, as well as at balancing around the desired set-point.

The equations of motion of the Pendubot may be formulated as follows:

$$\begin{cases} \dot{x} = f(x, u_R) \\ y = h(x) \end{cases} \quad (6)$$

Note that for the zero input, $u = 0$, the system (6) naturally exhibits in total four equilibrium positions, one stable and the other three unstable ones. From the control point of view, the most challenging are clearly the down-up ($\downarrow\uparrow$) and the up-up ($\uparrow\uparrow$) unstable equilibrium configurations. In the scope of this work is the latter one, i.e. transition from the resting $\downarrow\downarrow$ position to the unstable top $\uparrow\uparrow$ equilibrium by exploiting the system's nonlinear dynamics (6) subject to constraints.

The parameters such as the masses m_i , the links' lengths l_i , and the distance l_{ci} of the center of mass to the corresponding joint are directly measurable, and read as $m_1 = 0.265$ kg, $m_2 = 0.226$ kg, $l_1 = 0.206$ m, $l_2 = 0.298$ m, $l_{c1} = 0.107$ m and $l_{c2} = 0.133$ m. Even the moments of inertia I_1 and I_2 can be analytically determined accurately enough. The unknown friction model coefficients were chosen based on a number of off-line identification attempts yielding the following values: $c_1 = 0.477$, $c_2 = 2.098$, $c_3 = 0.649$, $c_4 = 0.040$, $c_5 = 0.727$, and $c_6 = 0.076$ to give the best fit in model verification.

III. CONTROLLER SYNTHESIS

This section introduces the formulation and implementation of the present control scheme based on combined nonlinear MHE and MPC for the fast Pendubot system. For clarity, we state the formulation first by deriving the estimation scheme.

A. MHE formulation

In many practical applications, the number of available measurements is typically smaller than the number of states. This is also the case for the Pendubot, where the dedicated sensors provide information about the angular positions of both links, leaving the respective angular velocities to be estimated. Recall also that unlike the another nonlinear estimation approaches, the MHE allows for the incorporation of constraints.

The nonlinear MHE state estimation problem is proposed to take the general form of a constrained least-squares (LSQ) dynamic optimization problem

$$\min_{x(\cdot), u(\cdot)} \frac{1}{2} \int_{t_0-T_E}^{t_0} (\|h(x) - \bar{y}\|_{Q_E}^2 + \|u - \bar{u}\|_{R_E}^2) dt, \quad (7a)$$

$$\text{s.t. } \frac{d}{dt}x = f(x, u), \quad (7b)$$

$$h_{\text{ineq}}(x, u) \leq 0. \quad (7c)$$

This nonlinear optimization problem is solved repeatedly at every time instant $t_k = kT_s$ ($k = 0, 1, \dots$), where T_s is the sampling time and T_E denotes the NMHE estimation horizon. The moving horizon objective function penalizes the deviation between the measurement model $h(\cdot)$ and the set of measurement data $\bar{y}$. The inclusion of the control variables within the objective is imposed to account for the noise collected during signal transfer, actuator inaccuracies and unmodeled dynamics. The appropriately chosen weighting matrices $Q_E \geq 0$, and $R_E > 0$ penalize the LSQ terms. The propagation of the system state is described by the equality (7b), whereas the upper and lower bounds for both states and input can be imposed via (7c). The output of the estimator is the initial state estimate ($\hat{x}$), which is then subsequently passed to the NMPC controller.

B. NMPC formulation

Similarly, the considered nonlinear MPC scheme is based on repeatedly solving the following optimal control problem (OCP) in effort of finding a function

$$u^*(\cdot) = \arg \min_{u(\cdot), x(\cdot)} \frac{1}{2} (\|x - x^{\text{ref}}\|_{P_C}^2)_{t=t_0+T_P} + \frac{1}{2} \int_{t_0}^{t_0+T_P} (\|x - x^{\text{ref}}\|_{Q_C}^2 + \|u - u^{\text{ref}}\|_{R_C}^2) dt, \quad (8a)$$

$$\text{s.t. } x(t_0) = \hat{x}(t_0), \quad (8b)$$

$$\frac{d}{dt}x = f(x, u), \quad (8c)$$

$$h_{\text{ineq}}(x, u) \leq 0 \quad (8d)$$

with T_P denoting the NMPC control horizon. Within this LSQ cost function, the deviation of the process control inputs u and states x from their reference trajectories, u^{ref} and x^{ref} , respectively, is penalized. The first summand is the so-called Mayer term, which evaluates the final costs raised

by the controlled variables at the given end time t_{k+T_P} . In this context usually referred to as the cost-to-go or terminal penalty function, the Mayer term is often included for stability reasons. As usual in tracking MPC applications, the norms in the objective function are weighted with penalty matrices $Q_C, P_C \geq 0$ and $R_C > 0$.

The system dynamics are described by the nonlinear right-hand side function $f(x, u)$ in equality (8c) with ordinary differential equations (ODEs). To obtain the current state x_k , we employ the aforementioned estimation scheme implied by the initial state constraint (8b). Vector $\hat{x}(t_k)$ is the system state estimated at time instant t_k by solving the nonlinear MHE problem (7). In this fashion, the NMPC problem (8) can be considered as a parametric nonlinear OCP. Similarly as in NMHE, the state and input constraints are imposed on the system by means of the inequality (8d).

C. Real-time implementation approach

By applying simultaneous approaches [4], [5], such as the direct multiple shooting [7], the infinite-dimensional optimal control or estimation problem are transformed to finite-dimensional ones. This way, the problems (7) and (8) are conveniently tackled with respect to unstable model dynamics (6). The arising discretized OCPs essentially lead to least-square nonlinear programming (NLP) problems. These can be efficiently solved using interior point or sequential quadratic approaches.

It is well known that the computational effort necessary to solve nonlinear optimal control and estimation problems exactly tends to easily become prohibitive, especially for systems with fast-evolving dynamics. To overcome this issue, only a single SQP iteration with Gauss-Newton Hessian approximation per sampling time is performed within the real-time iteration scheme [6]. Although solving the problems only approximately, its established efficacy is supported by employing the aforementioned simultaneous NLP parametrization, direct multiple-shooting method with/without condensing, and the so-called initial value embedding which constraints the initial value in the NLP to coincide with the estimated state of the system. This hence enables to perform most of the underlying computations before the measurement or current state estimate become available. Therefore, within the real-time iteration scheme, computations are divided into a longer and more demanding preparation phase, involving linearization, condensing etc., and a shorter feedback phase, where a single dense/sparse QP subproblem is solved.

The computational division into two aforementioned phases offers a possibility to parallelize particular algorithmic components when dealing with joint estimation and control problems. Considering a single-thread application, all the four phases are executed sequentially, starting with the feedback phase of NMHE (further referred to as estimation phase) and followed by the NMPC feedback phase once the state estimate is obtained. Since the order of the two consecutive preparation phases is not relevant, a decrease in the overall computation time can be achieved by parallelizing their execution, moreover, the NMHE preparation phase may be triggered simultaneously with the NMPC feedback phase. This idea is depicted in Fig. 3. For the purpose of parallelization, the openMP API was utilized in the real-time application.

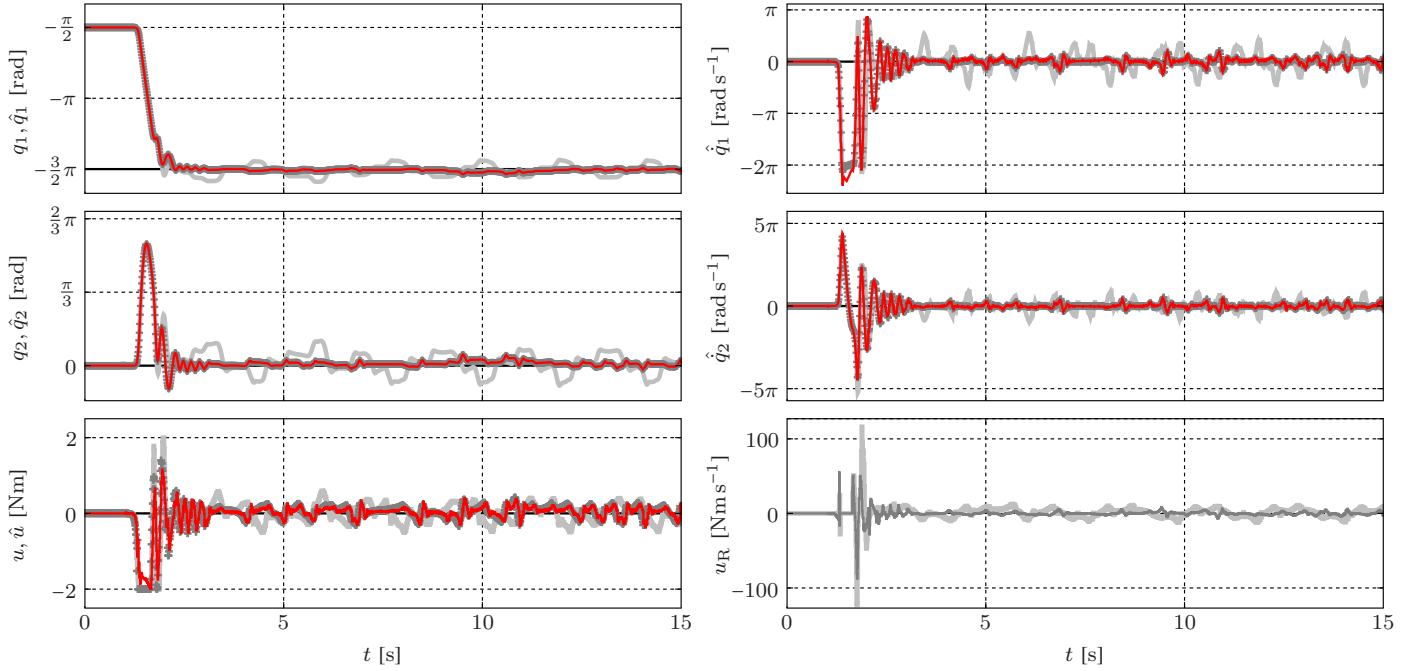


Fig. 2. Experimental results for the top ($\uparrow$) unstable equilibrium position: red line - system state and control input estimates, gray '+' markers - measurements, black line - references. The thick light gray curves in the background stand for the same control scenario, yet without considering the friction model.

We remark that both the nonlinear MHE and the NMPC formulations require to solve a whole sequence of "neighbouring" NLPs. In order to address this issue efficiently, subsequent problems are initialized based on the information from the previous ones. This approach is referred to as shift initialization [4]. For a detailed description, contractivity and stability proof of the RTI scheme we refer to [6], [10].

IV. EXPERIMENTAL RESULTS

In this section we put forward and analyze the experimental results obtained for the nonlinear Pendubot system using the on-line estimation-based predictive control scheme introduced in Section 3.

A. Hardware & software approach

The real-time experiments were performed on the real-life laboratory setup designed at the authors' workplace. For more details regarding the hardware setup we refer to [14].

As outlined in the introductory section, in order to numerically solve the optimization problems we make use of the ACADO Code Generation tool ([9]), exploiting direct multiple-shooting, RTI and sequential quadratic programming. To efficiently solve the underlying QP subproblems of the SQP-based RTI scheme, we make use of efficient quadratic programming solvers, in particular, the condensing-based qpOASES parametric solver implementing the on-line active set strategy proposed in [15], and the qpDUNES solver implementing a dual Newton strategy [16], which combines the warm-starting capabilities of active-set methods and the structure-exploiting features of interior-point methods.

The presented estimator-controller scheme is implemented on an ADLINK MXC-6321 embedded PC (Intel i7, 3.3 GHz,

12 GB RAM) featuring Ubuntu 14.04 with an RT preemptive real-time kernel as operating system.

B. Experimental setup

By applying the multiple shooting technique, the system dynamics given by the continuous ODE model (4)–(5) is parameterized using uniform intervals of $T_s = 10\text{ms}$. Gauss-Legendre Runge-Kutta integrator of order two for discretization over the shooting intervals. The estimation horizon was chosen as $T_E = 0.5$, while the prediction horizon was set as $T_C = 1\text{s}$. Moreover, at each sampling time the related nonlinear optimization problems are solved with the following bounds on states: $-2\text{Nm} \leq u \leq 2\text{Nm}$, $-2\pi\text{rad} \leq x_1 \leq 0\pi\text{rad}$. Within these box constraints, the ones imposed on the control input u stem from the internal limitations of the actuator. Similarly, bounds on the angular position of the actuated link x_1 are also imposed and motivated by preventing the sensor cable from twisting onto the rotor shaft.

The state references within the NMPC scheme are kept zero except of the ones for angular positions of both links, which are tuned for the desired top unstable equilibrium, i.e. $[\cos \frac{\pi}{2}, \sin \frac{\pi}{2}, 0, \cos 0, \sin 0, 0, 0]^T$ ¹, and $u_R^{\text{ref}} = 0$. The respective NMPC weighting matrices were chosen as $P_C = Q_C = \text{diag}([200, 200, 1, 100, 100, 1, 10])$, with input rate weight $R_C = 0.1$. Next, the MHE weights were selected as $Q_E = \text{diag}([3, 3])$ and $R_E = 0.001$. Note that the units of the weights are consistent with the variables in order to yield a dimensionless cost, and are omitted here for brevity.

¹This form of state and its associated reference vector was utilized within the NMPC objective to ease the reference angle tracking since each of the two links may, in theory, exhibit infinitely many possible angular values (with 2π period) for a given equilibrium.

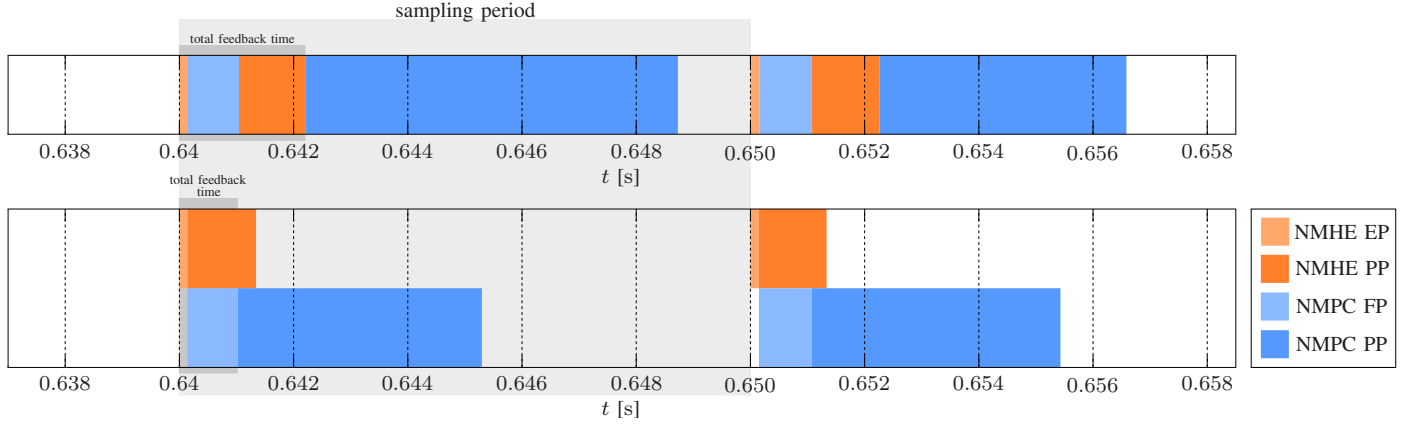


Fig. 3. Graphical illustration of the timing performance for the NMHE/NMPC RTI scheme's algorithmic components, within the standard (upper graph) and the parallelized (lower graph) configuration, taken from the real-time transition of the Pendubot between its stable and unstable ($\Downarrow \rightarrow \Uparrow$) equilibrium.

TABLE I. REAL-TIME PERFORMANCE ANALYSIS IN TERMS OF NMHE-NMPC ITERATIONS' TIMES, IN [ms], FOR VARIOUS CASE-STUDY CONFIGURATIONS AND CPU CLOCK FREQUENCIES.[†]

		sequential							with parallelization						
		NMHE			NMPC			total TET	NMHE			NMPC			total TET
		EP	PP	RTI	FP	PP	RTI		EP	PP	RTI	FP	PP	RTI	
@3.3 GHz	qpOASES	0.082	0.638	0.720	0.463	2.183	2.645	3.368	0.083	0.669	0.752	0.505	2.259	2.765	2.922
	$t_{\max}$	0.141	1.003	1.128	1.822	2.994	3.990	4.932	0.225	0.883	0.983	1.551	3.008	3.895	4.077
	qpDUNES	0.108	0.491	0.599	0.242	1.042	1.292	1.900	0.108	0.523	0.632	0.263	1.092	1.355	1.504
	$t_{\max}$	0.647	0.690	1.158	0.561	1.614	1.836	2.577	0.602	0.720	1.091	0.623	1.737	2.097	2.587
@1.6 GHz	qpOASES	0.150	1.258	1.409	0.926	4.419	5.341	6.757	0.165	1.301	1.466	0.910	4.445	5.356	5.599
	$t_{\max}$	0.340	1.904	2.076	3.011	6.520	7.520	9.018	0.458	1.762	2.112	2.760	5.111	7.258	7.463
	qpDUNES	0.207	1.030	1.258	0.449	2.120	2.571	3.756	0.200	0.984	1.184	0.502	2.146	2.646	2.880
	$t_{\max}$	1.110	1.347	2.050	0.834	2.715	3.203	4.970	0.347	1.586	1.906	0.741	3.561	4.341	4.632
@1.2 GHz	qpDUNES	0.260	1.334	1.594	0.581	2.886	3.467	5.064	0.253	1.311	1.564	0.626	2.928	3.555	3.853
	$t_{\max}$	0.890	2.141	2.628	1.383	4.456	5.151	7.541	1.096	1.732	2.337	1.617	4.344	5.260	5.532

[†] $t_{\text{avg}}/t_{\text{max}}$ - average/maximum computation time, total TET - overall joint task execution time of particular NMHE/NMPC iterations, EP/FP - estimation/feedback phase, PP - preparation phase, RTI - full real-time iteration, $N_E = 50$, $N_C = 100$.

To evaluate the computation performance of different hardware/software configurations, a series of experiments was conducted, where the control objective was to steer the Pendubot to its top unstable equilibrium. Results for one of the experiments are shown in Fig. 2. The control procedure itself is preceded by a initialization phase when the measurements are buffered in order to obtain proper estimator output. It can be observed that the state trajectories are rapidly driven and stabilized towards the desired setpoint. One may also notice the difference in the quality of the control when employing the nonlinear friction model (3), as compared with the results obtained with no friction model depicted as well. In the latter case, the controller exhibits an oscillating behaviour due to

friction causing a limit-cycle phenomenon around the set-point. By considering the friction model, the amplitude of the limit-cycle oscillations is suppressed, resulting in a much smoother stabilization and thus a better control performance.

Fig. 3 illustrates the timing performance of both sequential and parallel configurations taken from the equilibrium transition phase, in other words, the swing-up phase. After the new measurement $\bar{y}(k)$ is obtained at time $t = 0.64$ s the NMHE estimation phase is triggered. The estimation phase is followed by the NMPC feedback phase once the new state estimate $\hat{x}_0$ is available. The first member of the optimal control input vector u_0^* is immediately applied to the system

once it has been computed. The time elapsed from acquiring the measurement up to applying the input is referred to as total feedback time. In the parallel configuration, the NMHE preparation phase is independent from the NMPC procedures and thus may be executed simultaneously. This clearly leads to shortening the computation time providing the option to decrease the sampling period, which may be advantageous for controlling fast systems, or enables to successfully implement the estimator-control scheme on less powerful embedded multicore platforms.

To assess the computational complexity of the scheme, in Table I reports the execution times of its main algorithmic routines. For this purpose, we treat the iterations of the RTI scheme for both the NMHE and NMPC problem separately, and split the computation effort of each into a preparation phase and a feedback (estimation, in case of NMHE) phase. As outlined earlier, for solution of the nonlinear MHE/MPC optimization problems we utilize the condensing qpOASES-based approach as well as the sparsity exploiting qpDUNES-based approach. Both approaches clearly evidence that the RTI execution time is dominated by the preparation phase, where the effort is put mainly into linearization of the NLP. This is in particular significant for the former approach which additionally carries out a condensing routine resulting in the reduced-size QP. Overall, the remaining feedback/estimation phase is subsequently devoted to the solution of a single sparse/dense QP subproblem. The timing results indicate that the lower per-iteration complexity of the sparse QP strategy (see [16]) allows for computation times to scale better with the problem size, as compared to the condensing-based approach. To illustrate the solvers' performance on less powerful platforms, the CPU speed has been consecutively scaled down from 3.3 GHz to 1.6 GHz and finally 1.2 GHz. The worst-case computational effort over all real-time steps is reported as well. Inspecting the average and maximum computation times for different CPU clock frequencies, the two QP solvers and sequential/parallel configuration, one may conclude that the sparse qpDUNES solver is expectedly better suited for problems necessitating longer horizons. Still, the qpOASES solver performs quite well for the NMHE with the shorter horizon of 50 steps, however, fails when CPU is clocked at 1.2 GHz. Moreover, the performance within the real-time requirements can be further improved by exploiting the parallelization, as suggested by the considerably lower overall execution times leaving more headroom for other potential procedures and auxiliary tasks, eventually allowing for increasing the horizon lengths.

V. CONCLUSION

This paper proposed a real-time implementation of the combined NMHE-NMPC framework for the Pendubot control task, in particular achieving its challenging top unstable equilibrium. To synthesize the controller, we utilized the efficient RTI-based scheme exploiting the full nonlinear model, augmented with a nonlinear friction model. This approach allowed for a very good performance of the NMPC controller, to a great extent due to the necessary and highly reliable MHE estimates of unmeasured system states. The experimental verification of the proposed framework was moreover extended by an extensive performance analysis demonstrating the possible parallelization of certain algorithmic routines. The results

clearly suggest the potential applicability in terms of real-time deployment on embedded microcomputer-based platforms.

ACKNOWLEDGMENT

The authors gratefully acknowledge the contribution of the Slovak Research and Development Agency under the projects APVV-0090-10, APVV-0015-12 and APVV-14-0399, next, the Scientific Grant Agency of the Slovak Republic under the grant 1/0144/15, and the internal grant of the Slovak University of Technology for support of young researchers.

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